ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2024 - 2025)

LEADER & ACHIEVER COURSE

PHASE : MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO

TARGET : PRE MEDICAL 2025

Test Type : **MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 17-03-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	1	4	3	2	3	3	2	2	2	1	4	3	2	4	3	2	4	2	3	2	4	1	3	3	2	3	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	1	2	2	2	2	3	1	3	2	2	4	2	1	2	1	4	4	3	3	2	3	4	4	4	2	4	4	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	1	2	1	2	4	3	1	3	2	2	3	4	1	2	3	2	3	3	3	1	4	3	3	2	3	2	3	3	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	3	4	2	4	2	3	1	4	3	3	3	3	1	2	2	1	3	3	4	1	1	2	3	4	2	2	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	3	3	3	4	1	3	3	2	3	2	1	4	3	4	1	4	3	3	3	1	4	1	2	1	3	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	4	3	1	2	4	4	2	1	2	4	3	1	3	1	2	1	1	3	2	2	1	3	3	3	2	2	1	3	4

HINT – SHEET

10. **Ans (2)**

$$r_n = 0.529 \frac{n^2}{z} \text{ Å}$$

13. **Ans (4)**

5g subshell have 9 orbitals

14. **Ans (3)**

Magnetic moment = 1.73 BM

The no. of unpaired e^- = 1

$V = [Ar]\ 4s^2\ 3d^3$

$V^{+4} = [Ar]4s^0\ 3d^1$

In VCl_4 Oxidation state of 'V' is '+4'.

16. **Ans (4)**

fact

17. **Ans (3)**

	List-I Elements		List-II Atomic radii (pm)
(a)	O	(iii)	66 pm
(b)	C	(iv)	77 pm
(c)	B	(i)	88 pm
(d)	N	(ii)	74 pm

B > C > N > O

As we move across the period from left to right atomic radius decreases due to increase in nuclear charge.

18. Ans (2)

Factual theory based.

19. Ans (4)

$$\boxed{\text{I.P.} \propto Z_{eff} \propto \frac{\oplus\text{ve charge}}{\ominus\text{ve charge}}}$$

20. Ans (2)

Configuration of Y^+ & X^- = $2s^2\ 2p^6$ (noble gas) hence by gaining of an electron. X^- has noble gas configuration so electron affinity (EA) of X very high to become noble gas like. EA of X > EA of Y.

21. Ans (3)

Concept

22. Ans (2)

EA : Cl > O > C > N

ΔHeg_1 of N is positive

23. Ans (4)

$N_{(g)} + O^+_{(g)} \rightarrow N^+_{(g)} + O_{(g)}$

$N_{IP} > O_{IP}$

$N \longrightarrow N^+$ endo

24. Ans (1)

$N(g) + e^- \longrightarrow N^-(g) \quad \Delta_{eg}H \longrightarrow$ positive value

(energy absorbed)

$P(g) + e^- \longrightarrow P^-(g) \quad \Delta_{eg}H \longrightarrow$ negative value

(energy released)

25. Ans (3)

F has maximum electronegativity.

Cl has max. electron affinity.

26. Ans (3)

Acidic ch. $\propto$ oxidation state

27. Ans (2)

HE $F^- > Cl^- > Br^- > I^-$

$\therefore$ Electrical conductance

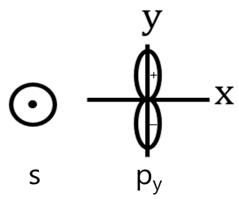
$F^- < Cl^- < Br^- < I^-$

28. Ans (3)

B.E $\Rightarrow Cl_2 > Br_2 > F_2 > I_2$

29. Ans (1)

S + P_y does not form any Bond if I.N.A. is 'x'



30. Ans (1)

O = O　　　N ≡ N

$2p_\pi - 2p_\pi$　　$2p_\pi - 2p_\pi$

S = S　　　P ≡ P

$3p_\pi - 3p_\pi$　　$3p_\pi - 3p_\pi$

Strength of overlapping $\longrightarrow 2p_\pi - 2p_\pi > 3p_\pi - 3p_\pi$

31. Ans (3)

Steric No. = σ + ℓ.P.

5 = 2 +(a = 3)

(b = 5) = 4 + 1

6 = 4 + (c = 2)

(d = 5) = 3 + 2

32. Ans (1)

```
    ..
H — P — H
    |
    H
```

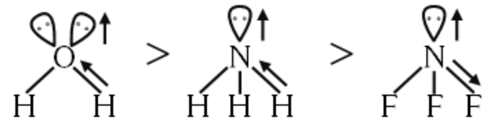
33. Ans (1)

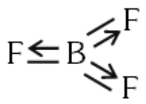
π bond strength $\uparrow\uparrow$ distance $\downarrow\downarrow$ (B.O.$\uparrow\uparrow$)

34. Ans (2)

$\mu = q \times d$

35. Ans (2)



 $(\mu = 0)$

36. Ans (2)

H_2O has intermolecular H-bonding

37. Ans (2)

Properties of H-bond. (X is covalent bond and F----H is H-bond)

38. Ans (3)

(P) Along the I.N.A. $\Rightarrow$ σ

(Q) p + d $\Rightarrow$ π bonding

(R) Opposite phase $\longrightarrow$ π antibonding

(S) Opposite phase $\longrightarrow$ σ bond (s–s)

39. Ans (1)

$O_2^+ = \sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma 2p_z^2$

$\left[\pi_{2p_x}^2 = \pi_{2p_y}^2\right] \cdot \left[\pi_{2p_x}^{*1} = \pi_{2p_y}^{*}\right]$

$O_2^- = \sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma 2p_z^2$

$\left[\pi_{2p_x}^2 = \pi_{2p_y}^2\right] \cdot \left[\pi_{2p_x}^{*2} = \pi_{2p_y}^{*1}\right]$

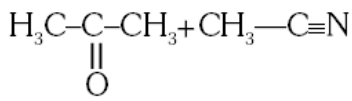
both have same number in BMO.

40. Ans (3)

Covalent character $\propto$ Polarisation.

41. Ans (2)

$KCl + H_2O$ $\Rightarrow$ ion-dipole attraction



dipole-dipole attraction

$HCl + Cl_2$ $\Rightarrow$ dipole induced dipole attraction

42. Ans (2)

$XeF_5^{\ominus}$ & SiF_6^{-2} exist

43. Ans (4)

Concept

45. Ans (1)

BE of $\longrightarrow$ O-O < S – S < S – H < O–H

46. Ans (2)

NCERT-XI, Pg. # 96, 97, 98, 100

47. Ans (1)

NCERT-XI, Pg. # 97, 98, 101

48. Ans (4)

NCERT (XI) Pg # 91

49. Ans (4)

NCERT XI Pg. # 98, 99, 100

50. Ans (3)

NCERT XI Page No. # 90, 96, 99

51. Ans (3)

NCERT XI page No. # 100

52. **Ans (2)**
NCERT XI Page. No :- 100

53. **Ans (3)**
NCERT-XI, Pg. 94

54. **Ans (4)**
NCERT-XI, Pg. 94

55. **Ans (4)**
NCERT (XI) Pg # 97

56. **Ans (4)**
NCERT (XI) Pg # 88

57. **Ans (2)**
NCERT-XI Pg # 136

58. **Ans (4)**
NCERT XI Pg # 102

59. **Ans (4)**
NCERT-XI, Pg. # 127

60. **Ans (1)**
NCERT, Pg. # 127

61. **Ans (3)**
NCERT-XI, Pg # 126, 127

62. **Ans (1)**
NCERT-XI, Pg. # 125

63. **Ans (2)**
NCERT-XI, Pg. # 127

64. **Ans (1)**
NCERT-XI, Pg. # 122

65. **Ans (2)**
NCERT (XI) Pg # 122

66. **Ans (4)**
NCERT-XI Pg. # 123

67. **Ans (3)**
NCERT-XI, Pg. # 123

68. **Ans (1)**
NCERT (XI) Pg # 122

69. **Ans (3)**
NCERT (XI) Pg # 126

70. **Ans (2)**
NCERT (XI) Pg # 121

71. **Ans (2)**
XI-NCERT Page No. # 126

72. **Ans (3)**
NCERT-XI, Pg. # 126 to 128

73. **Ans (4)**
NCERT XI Pg. # 121

74. **Ans (1)**
NCERT XI Pg : 111-112

75. **Ans (2)**
NCERT-XI, Pg. # 107

76. **Ans (3)**
NCERT-XI, Pg. # 110

77. **Ans (2)**
NCERT XI Pg. No. 106

78. **Ans (3)**
NCERT XI, Pg. No. # 109

79. **Ans (3)**
NCERT XI ; Page 108

80. **Ans (3)**
NCERT XI Pg. # 106

81. **Ans (1)**
NCERT Pg. # 107

82. **Ans (4)**
NCERT XI Pg. # 110

83. **Ans (3)**
NCERT XI Pg. # 108

84. **Ans (3)**
NCERT-XI, Pg # 113

85. **Ans (2)**
NCERT-XI, Pg # 117

86. **Ans (3)**
NCERT XI, Pg. # 112, 118

87. **Ans (2)**
NCERT XI Pg. # 118

88. **Ans (3)**
NCERT XI Pg. # 118

89. **Ans (3)**
NCERT Pg. # 118

90. **Ans (3)**
NCERT XI, Pg # 116

91. **Ans (1)**
NCERT Pg #281

92. **Ans (3)**
NCERT Pg #285

93. **Ans (3)**
NCERT Pg #287

94. **Ans (4)**
NCERT Pg #282

95. **Ans (2)**

	Neutrophils	>	Lymphocytes	>	Monocytes	>	Eosinophils	>	Basophils
Percentage of :	60 - 65%	>	20 - 25%	>	6 - 8%	>	2 - 3%	>	0.5 - 1%

96. **Ans (4)**
NCERT, Pg. # 194

97. **Ans (2)**
NCERT, Pg. # 280

98. **Ans (3)**
NCERT, Pg. # 280

99. **Ans (1)**
NCERT, Pg. # 196

100. **Ans (4)**
NCERT, Pg. # 197

101. **Ans (3)**
NCERT, Pg. # 200

102. **Ans (3)**
NCERT, Pg. # 202

103. **Ans (3)**
NCERT Pg # 279

104. **Ans (3)**
NCERT Page No. # 297

105. **Ans (1)**
NCERT Pg. # 194

106. **Ans (2)**
NCERT Pg. # 194

107. **Ans (2)**
NCERT Pg. # 195

110. **Ans (3)**
The correct answer is All living cells of the body. Here's why:Cellular Respiration: This is the process by which cells break down glucose (or other energy-rich molecules) to release energy in the form of ATP. This process occurs in the mitochondria of all living cells.

113. Ans (1)

CO_2 Sensitivity: The body's respiratory system is more sensitive to changes in CO_2

- levels than O_2 levels.

pH Change: As CO_2 accumulates in the blood, it reacts with water to form carbonic

- acid, which lowers the blood pH.

Chemoreceptor Activation: Specialized sensors called chemoreceptors detect this pH

- change and send signals to the brain.

Respiratory Center Stimulation: The brain's respiratory center receives these signals and

- initiates the urge to breathe.

While a decrease in O_2 levels can also contribute to the urge to breathe, it's typically a secondary factor. The body's primary mechanism for regulating breathing is through monitoring CO_2 levels.

114. Ans (2)

NCERT Pg. No. 185

115. Ans (3)

SOLUTION-Option 3: EC (Expiratory Capacity)

Explanation:

TV (Tidal Volume) is the amount of air inhaled or exhaled during normal, relaxed

- breathing.

ERV (Expiratory Reserve Volume) is the amount of air that can be forcibly exhaled

- after the normal tidal exhalation.

116. Ans (4)

NCERT PAGE NO. 190

117. Ans (2)

NCERT, Page#271

118. Ans (2)

NCERT Page no. 183

119. Ans (3)

NCERT (XI) Pg. # 171

120. Ans (4)

NCERT XI , Pg.# 275

121. Ans (3)

NCERT Pg. # 272

123. Ans (4)

NCERT Pg. no. 272

124. Ans (3)

NCERT Pg No#295 - 298

125. Ans (3)

NCERT Pg No#295 - 298

Following statements are true-

(I) Glomerular filtrate is isotonic to plasma.

(III) Filtrate is isotonic in Proximal Convoluted Tubule(PCT).

126. Ans (3)

NCERT Pg. No. 208

127. Ans (4)

NCERT XI Pg.# 293

128. Ans (1)

NCERT−XI, Para-6, Pg # 297 (E)

129. Ans (3)

NCERT−XI, Para−4, Pg # 298,

Para - (7)(H) Pg. # 298

130. Ans (3)

NCERT−XI, Pg. # 299

131. Ans (2)

NCERT XI Pg # 298

132. Ans (3)

NCERT-XI, Page # 211

133. Ans (2)

NCERT Page # 209

135. Ans (4)

NCERT−XI, Pg. # 290

136. Ans (3)

$$x_{cm} = \frac{2 \times 15 + 3 \times 0}{5} = 6$$

$$y_{cm} = \frac{2 \times 0 + 3 \times 20}{5} = 12$$

These values of x and y satisfy lines given is a, b and c.

137. Ans (4)

By COME

$$\frac{1}{2}\mu v_{rel}^2 = \frac{1}{2}kx^2$$

$$\frac{1}{2}\frac{(m)(m)}{m+m}(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$$

$$\frac{1}{2}\frac{m}{2}(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$$

$$\frac{1}{4}m(v_1 + v_2)^2 = \frac{1}{2}kx_{max}^2$$

138. Ans (1)

NCERT, Pg. # 99

$$\vec{v}_{cm} = \frac{1(t^2\hat{i}) + 1(t+1)\hat{j} + 1\left(\frac{3}{2}t\hat{i} + t^2\hat{j}\right)}{3}$$

at t = 2 sec

$$\vec{v}_{cm} = \frac{4\hat{i} + 3\hat{j} + 3\hat{i} + 4\hat{j}}{3}$$

$$\vec{v}_{cm} = \frac{7\hat{i} + 7\hat{j}}{3}$$

$$\therefore\ \vec{p}_{cm} = m\vec{v}_{cm}$$

$$= 3\left(\frac{7\hat{i} + 7\hat{j}}{3}\right) = (7\hat{i} + 7\hat{j})$$

$$\therefore\ p_{cm} = \sqrt{7^2 + 7^2} = 7\sqrt{2} \text{ units}$$

139. Ans (4)

NCERT, Pg. # 98

$\because$ COM will be at centroid

$$\therefore\ \frac{-2 - 4 + x}{3} = 0 \Rightarrow x = 6$$

$$\frac{3 - 2 + y}{3} = 0 \Rightarrow y = -1$$

140. Ans (3)

NCERT, Pg. # 100

$$v_{block} = \sqrt{2 \times 10 \times 0.2} = 2 \text{ m/s}$$

$$(mv)_{bullet} = (mv)_{block} + (mv')_{bullet}$$

$$5 = 4 + (mv')_{bullet}$$

$$1 = (0.01)\ v'_{bullet}$$

$$v'_{bullet} = \frac{1}{0.01} = 100 \text{ m/s}$$

141. Ans (3)

Velocity of ball 1 just after collision with mass 2m

$$v_1 = \frac{m-(1)(2m)}{m+2m}(0) + \frac{(1+1)(2m)}{m+2m}v_0$$

$$v_1 = \frac{4}{3}v_0$$

Now velocities between 1 and 2 will exchange because masses are equal and head-on elastic collision

$$\therefore\ v_2 = \frac{4}{3}v_0$$

Similarly $v_3 = v_2$ after collision and ball (2) comes at rest

$$\therefore v_3 = \frac{4}{3}v_0$$

142. Ans (3)

(a) $v_n = e^n\sqrt{2gh}$

$$\therefore\ v_2 = \left(\frac{1}{2}\right)^2\sqrt{2\times10\times180}$$

$$\therefore v_2 = 15 \text{ m/s}$$

(b) $v_{avg} = \sqrt{\frac{gh}{2}}\left(\frac{1+e^2}{(1+e)^2}\right)$

$$= \sqrt{\frac{10\times180}{2}}\left(\frac{1+1/4}{(3/2)^2}\right)$$

$$= (30)\left(\frac{5}{9}\right) = \frac{50}{3} \text{ m/s}$$

(c) Total distance

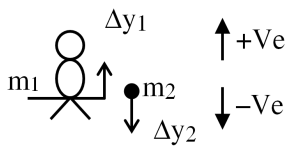
$$s = h\left(\frac{1+e^2}{1-e^2}\right)$$

$$= 180\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right) = 180\left(\frac{5/4}{3/4}\right)$$

$$= 300 \text{ m}$$

143. Ans (1)

No external force acting on (boy + ball) system, therefore displacement of CoM is zero.



$$\Delta y_{cm} = 0$$

$$m_1\Delta y_1 + m_2\Delta y_2 = 0$$

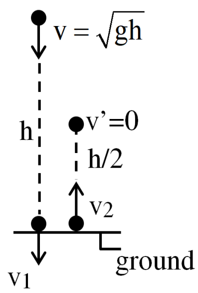
$$(40)\ \Delta y_1 = -(0.2)(-20)$$

$$\Delta y_1 = \frac{4}{40} = 0.1 \text{ m}$$

When ball reaches the floor, the distance of the man above the floor will be (20 + 0.1) = 20.1 m.

144. Ans (4)

$$v_1 = \sqrt{v^2 + 2gh} = \sqrt{3gh}$$



$$e = \frac{|v_2|}{|v_1|}$$

$$e = \frac{\sqrt{2g(h/2)}}{\sqrt{3gh}}$$

$$e = \frac{1}{\sqrt{3}}$$

145. Ans (1)

NCERT Pg # 41

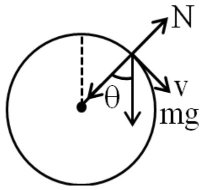
$$a_c = \frac{v^2}{R}$$

$$8 = \frac{v^2}{2} \Rightarrow v = 4$$

$$P = mv$$

$$= m\times4 = 4M$$

146. Ans (2)



$$mg\cos\theta - N = \frac{mv^2}{R}$$

$$N = mg\cos\theta - \frac{mv^2}{R}$$

147. Ans (1)

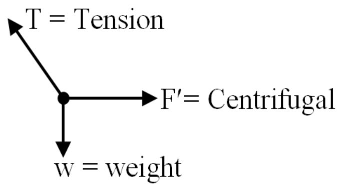
NCERT Pg # 64

$$V_{max} = \sqrt{\mu_s Rg} = \sqrt{\frac{1}{10} \times 3 \times 9.8} = 1.71 \text{ m/s}$$

$$V = 10 \times \frac{5}{18} = 2.7 \text{ m/s}$$

$V > V_{max}$

148. Ans (3)



149. Ans (2)

$$T_B = \frac{mV_B^2}{R} - mg = \frac{m\,5gR}{R} - mg$$
$$= 4\text{ mg}$$

150. Ans (2)

NCERT Pg. # 41

In U.C.M. speed is constant and acceleration is changing in direction.

151. Ans (2)

NCERT Pg. # 41

$$a_T = \frac{dV}{dt} = 2t = 4$$

$$a_C = \frac{v^2}{R} = \frac{(t^2)^2}{R} = \frac{(4)^2}{2} = 8$$

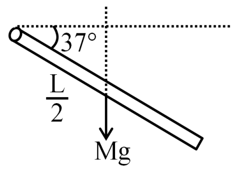
$$a_{net} = \sqrt{a_C^2 + a_T^2} = \sqrt{80}$$

152. Ans (4)

NCERT Pg.#106

Torque is axial vector

153. Ans (3)



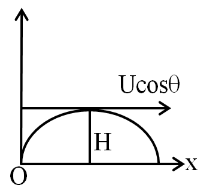
$$\tau = I\alpha$$

$$Mg\,\frac{L}{2}\,\frac{4}{5} = \frac{ML^2}{3}\alpha$$

$$\frac{6g}{5L} = \alpha$$

154. Ans (1)

NCERT Pg. # 106



$$L = mV\,r_{\perp ar}$$

$$L = \frac{mU\cos\theta U^2\sin^2\theta}{2g}$$

155. Ans (2)

NCERT Pg.#116

$$\frac{1}{3}M(2\ell)^2 = MK^2$$

$$K = \frac{2\ell}{\sqrt{3}}$$

156. Ans (4)

$$I_{AB} = I_{CM} + Md^2$$

$$I_A = mR^2 + mR^2$$

$$I_A = 2mR^2$$

$$2\pi R = \ell \Rightarrow R = \frac{\ell}{2\pi}$$

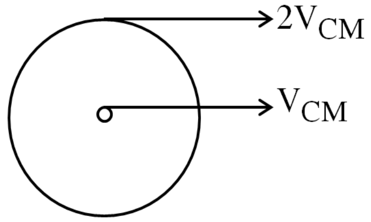
$$I_A = 2m\left\{\frac{\ell}{\pi}\right\}^2 = \frac{m\ell^2}{2\pi^2}$$

157. Ans (4)

$\tau = \frac{dL}{dt} = 0$

L = Const.

158. Ans (2)



160. Ans (2)

$I_{AB} = I_{CM} + Md^2$

161. Ans (4)

$\nu = \sqrt{\frac{2gh}{1+\frac{k^2}{R^2}}} = \sqrt{\frac{2gh}{1+\frac{2}{5}}}$

$\nu = \sqrt{\frac{10gh}{7}}$

162. Ans (3)

$\frac{3}{2}MR^2\alpha = f(2R)$

$\alpha = \frac{4f}{3MR}$

$\alpha R = \frac{4f}{3M}$

163. Ans (1)

$g = \frac{GMe}{Re^2} = \frac{G}{Re^2}\rho\frac{4}{3}\pi Re^3 = \frac{4}{3}G\pi\rho\, Re$

$\frac{\Delta g}{g} \times 100 = \left(\frac{\Delta\rho}{\rho} + \frac{\Delta Re}{Re}\right)100$

= (3 + 1)%

= 4%

164. Ans (3)

$I_{inside} = 0$

F = mI

F = 0

165. Ans (1)

$g_h = \frac{g}{\left(1+\frac{h}{R}\right)^2} = \frac{g}{\left(1+\frac{R/2}{R}\right)^2} = \frac{4}{9}g$

$W' = mgh = m \times \frac{4}{9} \times g = \frac{4}{9}W = \frac{4}{9} \times 36$

W' = 16 N

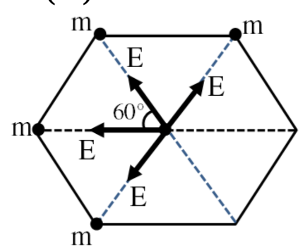
166. Ans (2)

Gravitational force provide centripetal force

$\frac{Gm^2}{4R^2} = \frac{mv^2}{R}$

$\boxed{v = \sqrt{\frac{Gm}{4R}}}$

167. Ans (1)



$E_{net} = \frac{\sqrt{3}Gm}{a^2}$

168. Ans (1)

$V_e \propto R_e\sqrt{\rho}$

169. Ans (3)

$W = U_f - U_i$

$W = -\frac{3Gm^2}{\ell}$

170. Ans (2)

$E = -\frac{dv}{dx} = -(\text{slope})$

171. Ans (2)

$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$

$\vec{E} = -k\left[4x\hat{i} - 2y\hat{j} + 2z\hat{k}\right]$

At (1,1,1)

$\left|\vec{E}\right| = k\sqrt{16+4+4}$

$= 2k\sqrt{6}$

172. Ans (1)

Let in water immersed volume is v_1 then

$w = (Th)_o + (Th)_w$

$V\rho_B g = (V-V_1)\ \rho_{og} + V_1\ \rho_w g$

$V \times 0.9\times10^3 = (V-V_1) \times 0.8 \times10^3 + V_1 \times 10^3$

$V(0.9\times10^3 - 0.8\times10^3) = V_1\ (10^3 - 0.8\times10^3)$

$\frac{V_1}{V} = \frac{1}{2}$

173. Ans (3)

$R = n^{1/3}r = (64)^{1/3}r$

$R = 4r$

$V_T \propto r^2$

$$\frac{(V_T)_{big}}{(V_T)_{small}} = \left(\frac{R}{r}\right)^2 = \left(\frac{4r}{r}\right)^2 = 16$$

$(V_T)_{big} = 16 \times 1.5 = 24\ ms^{-1}$

174. Ans (3)

For $R_{max} \Rightarrow h = \frac{H}{2}$

$v = \sqrt{2gh} = \sqrt{gH}$

$R_{max} = v\sqrt{\frac{2h}{g}} = \sqrt{gH}\sqrt{\frac{2H}{2g}}$

$H = 10$ m

175. Ans (3)

For $\theta = 90°$ then in capillary liquid no rise and no fall.

176. Ans (2)

$\frac{E}{V} = \frac{1}{2}Y\ .\ Strain^2$

177. Ans (2)

$W = T \times \Delta A$

$= T \times [2(SA_f - SA_i)]$

$$T = \frac{W}{2(SA_f - SA_i)} = \frac{2\times10^{-4}}{2(10\times6 - 8\times3.75)\times10^{-4}}$$

$= 3.3 \times 10^{-2}$ N/m

178. Ans (1)

Equation of continuity,

$a_1v_1 = a_2\ v_2$

$\pi r_1^{\ 2} \times v_1 = \pi r_2^{\ 2} \times v_2$

$$r_2 = \sqrt{\frac{r_1^2 v_1}{v_2}} = \sqrt{\frac{10\times10\times1}{4}} = 5cm.$$

179. Ans (3)

Pull or attraction by a train is application of bernoullis theorem.

180. Ans (4)

Given,

Pressure at point 1 $P_1 = 4000$ Pa

Pressure Velocity $V_1 = 4$ m/s

Velocity $V_2 = 2$ m/s

$\Rightarrow\ \rho = 800\ kg/m^3$

Using Bernoullis equation -

$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$\Rightarrow 4000 + \frac{1}{2}800(4)^2 = P_2 + \frac{1}{2}(800)(2)^2$

$\Rightarrow 4000 + 6400 = P_2 + 1600$

$P_2 = 10{,}400 - 1600$

$\Rightarrow$ 8800 Pa